

Alireza Salemi

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RESEARCH SUMMARY

PhD candidate and Google PhD Fellow working at the intersection of information retrieval and large language models, with research experience at Google DeepMind and Google Cloud AI.

Interests: RAG · LLM personalization · Post-training · Alignment · Agentic systems · Reasoning

EDUCATION

University of Massachusetts Amherst

Sep 2022 – Present

PhD, Computer Science (MSc 2024) · GPA 3.93/4.0 · advised by Hamed Zamani

- Awarded the **Google PhD Fellowship 2025** in Natural Language Processing.
- Two **SIGIR best-paper awards**: Best Short Paper (2024) and Best Student Paper (2023).
- **2nd place** (\$30K) at the 2026 Grounded Reasoning Cup, an AI-agent competition sponsored by Google, OpenAI, and Anthropic, ahead of teams from top universities.

University of Tehran

Sep 2017 – Feb 2022

BSc, Computer Engineering · GPA 19.60/20 (ranked 1st of 103) · advised by Azadeh Shakery

RESEARCH EXPERIENCE

Student Researcher — Google Cloud AI

Jun 2025 – Dec 2025

Hosts: Hamid Palangi and Mihir Parmar

- Designed blackboard-style multi-agent systems for autonomous information discovery in data-science workflows; 13–57% relative gains over the strongest baselines.

Student Researcher — Google DeepMind

Jun 2024 – May 2025

Hosts: Cheng Li, Mingyang Zhang, and Michael Bendersky

- Personalized and aligned LLMs with reinforcement learning, reinforced self-training, inference-compute scaling, reasoning, and pluralistic alignment (REST-PG, Pathways of Thoughts).

Data Science Intern — Lowe's

May 2023 – Aug 2023

Host: Surya Kallumadi

- Optimized dense retrievers that feed personal documents to LLMs via reinforcement learning and knowledge distillation from the model's own feedback, improving LLM personalization.

Research Assistant — Center for Intelligent Information Retrieval

Sep 2022 – Present

Advisor: Hamed Zamani

- Research on RAG, LLM personalization, and agentic systems—building benchmarks, reinforcement-learning and self-training methods, and evaluation frameworks grounded in retrieved evidence.

NLP Data Scientist — Sotoon

Nov 2021 – Aug 2022

Team lead: Hakimeh Fadaei

- Built an AutoML system that selects models and hyperparameters for text classification, powering sentiment analysis and automatic moderation of user-generated product reviews.

PUBLICATIONS

📄 1,722 citations | h-index 16 | i10-index 18 | *Google Scholar, July 2026*

SELECTED PUBLICATIONS

LaMP: When Large Language Models Meet Personalization

ACL 2024

Alireza Salemi, Sheshera Mysore, Michael Bendersky, Hamed Zamani

[Paper](#) · [Code](#)

First benchmark for LLM personalization (7 tasks); 590+ citations and a standard reference for the field.

Evaluating Retrieval Quality in Retrieval-Augmented Generation	SIGIR 2024
Alireza Salemi, Hamed Zamani 🏆 Best Short Paper	Paper · Code
<i>eRAG scores each retrieved document by its downstream contribution—Kendall’s τ +0.17–0.49 at up to 50× lower GPU cost.</i>	
A Symmetric Dual Encoding Dense Retrieval Framework for Knowledge-Intensive Visual Question Answering	SIGIR 2023
Alireza Salemi, Juan Altmayer Pizzorno, Hamed Zamani 🏆 Best Student Paper	Paper · Code
<i>DEDR encodes queries and documents in a shared space; +11.6–30.9% retrieval and state-of-the-art end-to-end VQA.</i>	
LLM-based Multi-Agent Blackboard System for Information Discovery in Data Science	Under review (NeurIPS 2026)
Alireza Salemi, Mihir Parmar, Palash Goyal, Yiwen Song, Jinsung Yoon, Hamed Zamani, Hamid Palangi, Tomas Pfister	Paper
<i>Specialized agents self-select tasks via a shared blackboard—no rigid hierarchy; 13–57% relative gains. (Google Cloud AI.)</i>	
Optimization Methods for Personalizing Large Language Models through Retrieval Augmentation	SIGIR 2024
Alireza Salemi, Surya Kallumadi, Hamed Zamani	Paper · Code
<i>First to optimize the retriever that feeds personal documents to an LLM via RL and knowledge distillation; significant gains on 6/7 LaMP tasks. (Lowe’s.)</i>	
GrepSeek: Training Search Agents for Direct Corpus Interaction	To be submitted (ICLR 2027)
Alireza Salemi, Chang Zeng, Atharva Nijasure, Jui-Hui Chung, Razieh Rahimi, Fernando Diaz, Hamed Zamani	Paper · Code
<i>Trains LLM agents to search corpora via shell commands using verified-trajectory generation + policy optimization (RL); strong gains across seven QA benchmarks.</i>	
OTHER PUBLICATIONS	
Critic-R: Improving Agentic Search using Instruction-tuned Retrievers with Natural Language Introspective Feedback	Under review (EMNLP 2026)
Md Zarif Ul Alam, Alireza Salemi, Hamed Zamani	Paper
Improving User Privacy in Personalized Generation: Client-Side Retrieval-Augmented Modification of Server-Side Generated Speculations	arXiv 2026
Alireza Salemi, Hamed Zamani	Paper · Code
Evaluation of Agents under Simulated AI Marketplace Dynamics	SIGIR 2026
To Eun Kim, Alireza Salemi, Hamed Zamani, Fernando Diaz	Paper
Learning to Reason for Multi-Step Retrieval of Personal Context in Personalized Question Answering	SIGIR 2026
Maryam Amirizani, Alireza Salemi, Hamed Zamani	Paper · Code
Learning from Natural Language Feedback for Personalized Question Answering	SIGIR 2026
Alireza Salemi, Hamed Zamani	Paper
Beyond a Million Tokens: Benchmarking and Enhancing Long-Term Memory in LLMs	ICLR 2026
Mohammad Tavakoli, Alireza Salemi, Carrie Ye, Mohamed Abdalla, Hamed Zamani, J. Ross Mitchell	Paper
Pathways of Thoughts: Multi-Directional Thinking for Long-Form Personalized Question Answering	WWW 2026
Alireza Salemi, Cheng Li, Mingyang Zhang, Qiaozhu Mei, Zhuowan Li, Spurthi Amba Hombaiah, Weize Kong, Tao Chen, Hamed Zamani, Michael Bendersky	Paper
Plan-and-Refine: Diverse and Comprehensive Retrieval-Augmented Generation	ICTIR 2026
Alireza Salemi, Chris Samarinas, Hamed Zamani	Paper · Code
LaMP-QA: A Benchmark for Personalized Long-Form Question Answering	EMNLP 2025
Alireza Salemi, Hamed Zamani	Paper · Code

Personalized Generation in the Large Model Era: A Survey	ACL 2025
Yiyan Xu, Jinghao Zhang, Alireza Salemi, Xinting Hu, Wenjie Wang, Fuli Feng, Hamed Zamani, Xiangnan He, Tat-Seng Chua	Paper
ExPerT: Effective and Explainable Evaluation of Personalized Long-Form Text Generation	Findings of ACL 2025
Alireza Salemi, Julian Killingback, Hamed Zamani	Paper · Code
Beyond Factual Accuracy: Evaluating Coverage of Diverse Factual Information in Long-Form Text Generation	Findings of ACL 2025
Chris Samarin, Alexander Krubner, Alireza Salemi, Youngwoo Kim, Hamed Zamani	Paper · Code
Accelerating Retrieval-Augmented Generation	ASPLOS 2025
Derrick Quinn, Mohammad Nouri, Neel Patel, John Salihu, Alireza Salemi, Sukhan Lee, Hamed Zamani, Mohammad Alian	Paper
Reasoning-Enhanced Self-Training for Long-Form Personalized Text Generation	arXiv 2025
Alireza Salemi, Cheng Li, Mingyang Zhang, Qiaozhu Mei, Weize Kong, Tao Chen, Zhuowan Li, Michael Bendersky, Hamed Zamani	Paper
Learning to Rank for Multiple Retrieval-Augmented Models through Iterative Utility Maximization	ICTIR 2025
Alireza Salemi, Hamed Zamani	Paper
Comparing Retrieval-Augmentation and Parameter-Efficient Fine-Tuning for Privacy-Preserving Personalization of Large Language Models	ICTIR 2025
Alireza Salemi, Hamed Zamani	Paper · Code
CIIR@LiveRAG 2025: Optimizing Multi-Agent Retrieval-Augmented Generation through Self-Training	LiveRAG @ SIGIR 2025
Alireza Salemi, Mukta Maddipatla, Hamed Zamani	Paper · Code
Towards a Search Engine for Machines: Unified Ranking for Multiple Retrieval-Augmented Large Language Models	SIGIR 2024
Alireza Salemi, Hamed Zamani	Paper
LongLaMP: A Benchmark for Personalized Long-Form Text Generation	Under review (TKDD)
Ishita Kumar, Snigdha Viswanathan, Sushrita Yerra, Alireza Salemi, Ryan Rossi, Franck Dernoncourt, Hanieh Deilamsalehy, Xiang Chen, Ruiyi Zhang, Shubham Agarwal, Nedim Lipka, Hamed Zamani	Paper
Retrieval-Enhanced Machine Learning: Synthesis and Opportunities	Under review (IR Journal)
To Eun Kim, Alireza Salemi, Andrew Drozdov, Fernando Diaz, Hamed Zamani	Paper
Pre-Training Multi-Modal Dense Retrievers for Outside-Knowledge Visual Question Answering	ICTIR 2023
Alireza Salemi, Mahta Rafiee, Hamed Zamani	Paper
PEACH: Pre-Training Sequence-to-Sequence Multilingual Models for Translation with Semi-Supervised Pseudo-Parallel Document Generation	LoResMT @ EACL 2023
Alireza Salemi, Amirhossein Abaskohi, Sara Tavakoli, Yadollah Yaghoobzadeh, Azadeh Shakery	Paper · Code
ARMAN: Pre-training with Semantically Selecting and Reordering of Sentences for Persian Abstractive Summarization	EMNLP 2021
Alireza Salemi, Emad Kebriaei, Ghazal Neisi Minaei, Azadeh Shakery	Paper · Code
A Comparative Analysis of Toxic Span Detection using Attention-based, Named Entity Recognition, and Ensemble Models	SemEval @ ACL 2021
Alireza Salemi, Nazanin Sabri, Emad Kebriaei, Behnam Bahrak, Azadeh Shakery	Paper · Code

HONORS AND AWARDS

Grounded Reasoning Cup — 2nd Place and \$30K prize: live AI-agent competition on retrieval and reasoning over enterprise documents (Data + AI Summit 2026); sponsored by Google, OpenAI, and Anthropic; finished ahead of teams from leading universities.

Google PhD Fellowship in Natural Language Processing: awarded to exceptional PhD students with outstanding research potential; proposal on collaborative optimization of agents.	2025
PhD Candidacy with Distinction: for exceptional performance in the PhD comprehensive exam.	2025
SIGIR 2024 Best Short Paper Award: Evaluating Retrieval Quality in Retrieval-Augmented Generation.	2024
SIGIR 2023 Best Student Paper Award: A Symmetric Dual Encoding Dense Retrieval Framework for Knowledge-Intensive Visual Question Answering.	2023
W. Bruce Croft Graduate Scholarship: for research in information retrieval and NLP.	2023, 2025
UMass Amherst CICS Scholarship: merit-based award for selected entering graduate students.	2022
Best Thesis Award, Computer Engineering: University of Tehran — BSc thesis selected as the best in the Computer Engineering graduating class for its research contribution.	2021
University of Tehran top-student awards: UT Sponsors Foundation Top Student, UT Scholarship, and Faculty of Engineering Award (top of 103 students).	2017–2020

INVITED TALKS

Personalization in the LLMs Era — Bocconi University	May 2026
Personalization in the LLMs Era — Carnegie Mellon University	Oct 2025
Personalization in the LLMs Era — Google DeepMind	May 2025
Personalization in the LLMs Era — UMass Amherst	Apr 2025
Knowledge-Intensive Visual Question Answering — QIAU	Dec 2023
When Large Language Models Meet Personalization — UMass ML Club	Oct 2023
When Large Language Models Meet Personalization — Google Research	May 2023

PROFESSIONAL SERVICE

Organizing

- **Workshop organizer**, Second Workshop on Customizable NLP (CustomNLP4U) — ACL 2026.
- **Tutorial organizer**, Retrieval-Enhanced Machine Learning — SIGIR 2025 and SIGIR-AP 2024.

Program Committee and Reviewing

- **Program committee and reviewer** (80+ papers): NeurIPS, ACL, EMNLP, SIGIR, WWW, CIKM, ICTIR, SIGIR-AP, LLM-IGS; journals ACM TOIS and IEEE TKDE.

Mentoring

- Independent study — mentored 15+ students across RAG, personalization, and agents.
- Industry Mentorship Program — MS students on LLM personalization, with Adobe mentors.
- Early Research Scholar Program — undergraduates on QA and LLM personalization.